Transfer Learning for Phosphene-Based Object Recognition in Cortical Visual Prostheses

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Abstract

Cortical visual prostheses work by stimulating the visual cortex directly, bypassing damaged optical pathways. The catch is resolution: a healthy retina has around a million photoreceptors, while current implants offer between 60 and 3,000 electrodes. What users see is not a picture but a sparse field of dots—phosphenes—that must be interpreted with training and effort. This paper asks whether a standard convolutional network, pre-trained on natural photographs, can still recognise objects once those photographs have been converted into phosphene-like dot patterns. We built a six-stage simulation pipeline, applied it to CIFAR-10, and compared EfficientNet-B0 fine-tuned from ImageNet weights against the same network trained from random initialisation. After ten epochs, the pre-trained model reached 69.21% accuracy on the test set versus 62.31% for the from-scratch model. The gap was sharpest at epoch 1 (20.67 points), which is hard to explain by anything other than the pre-trained weights already encoding structural features that survive phosphene degradation. Per-class numbers show the advantage holds across all ten categories, with cats benefiting most (+14.80 points)—an animal where coarse silhouette alone is not enough for reliable recognition.

Index Terms—cortical visual prosthesis, bionic eye, phosphene simulation, transfer learning, EfficientNet, deep learning, object recognition.

I. INTRODUCTION

For people who have lost sight entirely—not from retinal damage but from injury or disease further along the visual pathway—retinal implants are not an option. Cortical prostheses offer a different route: skip the eye and retina altogether and stimulate the visual cortex directly. Devices pursuing this approach include the Orion system from Cortigent, the Intracortical Visual Prosthesis (ICVP) developed at the Illinois Institute of Technology, and Neuralink's Blindsight, which received FDA Breakthrough Device designation in 2024 [1]–[3]. All of them are in human testing at various stages.

The electrode count problem defines what these devices can and cannot do. A million photoreceptors against at most a few thousand electrodes means that whatever the implant delivers, it is not a picture. Each active electrode produces a phosphene: a small blob of perceived light whose location, size, and brightness depend on stimulation parameters and where that electrode sits on the cortex. Put several phosphenes together and you get an approximate sketch of a scene—sparse, noisy, and requiring sustained effort to interpret [4].

One way to help is to put a smart image-processing layer between the camera and the electrode array. If the device can work out what it is looking at, it can send the user a useful summary rather than a raw dot map. That raises a practical question for anyone designing such a layer: should the recognition model be trained from scratch on phosphene-like data, or is it worth starting from ImageNet pre-trained weights? Transfer learning is the default answer in most computer vision settings [5], [6], but phosphene images are so degraded that the usual assumption—that natural-image features will transfer—needs checking.

This paper reports a controlled experiment addressing that question. We describe a six-stage phosphene simulation pipeline, apply it to CIFAR-10, and train EfficientNet-B0 under two conditions: starting from ImageNet weights and starting from random initialisation. We report overall accuracy, epoch-by-epoch learning curves, and per-class breakdowns. The experiment is deliberately simple—one architecture, one dataset, ten epochs—because the goal is a clean comparison, not a production system.

II. BACKGROUND AND RELATED WORK

A. Cortical Visual Prostheses

Retinal implants need a functioning retina to work, which rules them out for patients with optic nerve damage or cortical injury. Cortical prostheses go around that problem by stimulating V1 directly. Brindley and Lewin reported the first cortical stimulation experiments in 1968 [7]; sixty years on, the hardware has changed but the core idea has not. The ICVP currently uses a wireless 400-electrode array that has run continuously in human volunteers for over two years [3]. Neuralink's Blindsight aims for approximately 3,000 electrodes across a wider patch of cortex [2]. The Cortigent Orion sits at the lower end with 60 electrodes but has the longest track record in human use [1].

B. Phosphene Perception and Simulation

Cortical phosphenes are not uniform dots. Their shape, size, and brightness depend on electrode depth, current waveform, and the local cortical structure, and they drift with prolonged stimulation [4], [8]. Computational models—most notably the pulse2percept framework [9]—can predict what different electrode configurations will produce for a given patient. Our simulation is simpler: we are not trying to model any specific implant but to produce images that share the key statistical properties of phosphene fields (low resolution, binary-ish contrast, sparse structure) in a controlled way.

C. Deep Learning in Prosthetic Vision

Martel et al. [10] proposed putting a scene-understanding network upstream of the electrode encoder so the device could summarise what it sees rather than transmitting raw pixel data. Separate work has used deep neural networks as computational models of patient perception, sidestepping the cost and logistics of human trials [9]. Vision Transformers have been studied as alternative encoders [15]. What has not been reported, to our knowledge, is a direct comparison of transfer learning versus random initialisation on phosphene-simulated input under a standard classification benchmark.

D. Transfer Learning

Yosinski et al. [5] showed that ImageNet features—especially in early layers—transfer well across visual tasks, though the benefit shrinks for deeper, more task-specific layers. That work was done on natural images with moderate domain shift. Whether the same holds when the target images have been reduced to binary dot patterns is a different question. EfficientNet [12] scales depth, width, and resolution jointly under a compound coefficient derived empirically. The B0 variant has around 5.3 million parameters and runs fast enough to be plausible on wearable hardware, which motivated our choice of architecture.

III. METHODOLOGY

A. Phosphene Simulation Pipeline

Each image goes through six steps before it reaches the network. Together they strip out colour, fine texture, and mid-frequency detail, leaving a sparse binary pattern that roughly resembles what a cortical implant user reports seeing:

- **Grayscale conversion** — RGB collapsed to luminance using ITU-R BT.601 coefficients.
- **Gaussian blur** — $\sigma = 2.0$ pixels, mimicking the lateral spread of current from a single electrode.
- **Downsampling** — resized to 64×64, one cell per notional stimulation site.
- **Binary thresholding** — pixels above a fixed threshold go to full brightness, the rest go to black.
- **Noise injection** — 10% of pixels are randomly zeroed to simulate electrode failures.
- **RGB replication** — the single channel is copied to three channels for model compatibility.

Figure 1 shows what the pipeline does to a sample of CIFAR-10 images. The loss of detail is severe—in the harder categories like cats and birds, the original and simulated images look almost unrelated.

Fig. 1. CIFAR-10 originals (left of each pair) and their phosphene-simulated versions (right), produced by the six-stage pipeline in Section III-A. Colour, texture, and fine spatial structure are gone; what remains is coarse binary shape.

B. Dataset

We used CIFAR-10 [13]: 50,000 training images and 10,000 test images across ten classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck), all 32×32 pixels. Images were resized to 224×224 for EfficientNet-B0. ImageNet channel normalisation ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$) was applied in both training conditions so that normalisation is not a confound.

C. Model Architecture

EfficientNet-B0 [12] was used as-is except for the classification head, which we replaced with a linear layer mapping the 1,280-dimensional output to ten logits. All layers were fine-tuned in the transfer-learning condition—none were frozen. We did not freeze any layers because the phosphene domain is different enough from natural images that the intermediate representations likely need adjusting too, not just the head.

D. Training Protocol

Both models used AdamW [18] with a starting learning rate of 1×10^{-4} and cosine annealing, cross-entropy loss, batch size 32, and ten epochs. The random seed was fixed before each run. No augmentation was used beyond the phosphene simulation itself; adding augmentation would have made it harder to tell whether differences came from weight initialisation or data processing.

TABLE I TRAINING HYPERPARAMETERS

Hyperparameter	Value
Framework	PyTorch
Optimiser	AdamW
Learning rate	1×10^{-4}
LR schedule	Cosine annealing
Batch size	32
Epochs	10
Loss function	Cross-entropy
Phosphene grid	64×64

IV. RESULTS

A. Overall Accuracy and Loss

After ten epochs, the transfer-learning model reached 69.21% test accuracy (loss: 0.9387) and the from-scratch model reached 62.31% (loss: 1.0837). A 6.90-point gap on images that are barely recognisable even to a human is a non-trivial result. Table II gives the summary.

TABLE II FINAL-EPOCH PERFORMANCE COMPARISON

Model	Train Acc.	Test Acc.	Test Loss
Transfer Learning (ImageNet)	79.03%	69.21%	0.9387
From Scratch (random init.)	62.99%	62.31%	1.0837
Improvement	+16.04 pp	+6.90 pp	−0.1450

B. Training Dynamics

Figure 2 plots loss and accuracy across all ten epochs. The most striking number is at epoch 1: the pre-trained model is already at 60.92% while the from-scratch model is at 40.25%. That 20.67-point gap before either model has done much fine-tuning is what you would expect if pre-trained weights arrived already encoding features that work on phosphene data. After that, the gap closes steadily as the from-scratch model catches up, reaching 6.90 points by epoch 10.

Both models plateau between epochs 8 and 10 on the test set, so ten epochs is a reasonable stopping point. The training accuracy keeps rising for the pre-trained model (50% at epoch 1 to 79% at epoch 10), which suggests some overfitting that a larger dataset or regularisation strategy could address in future work.

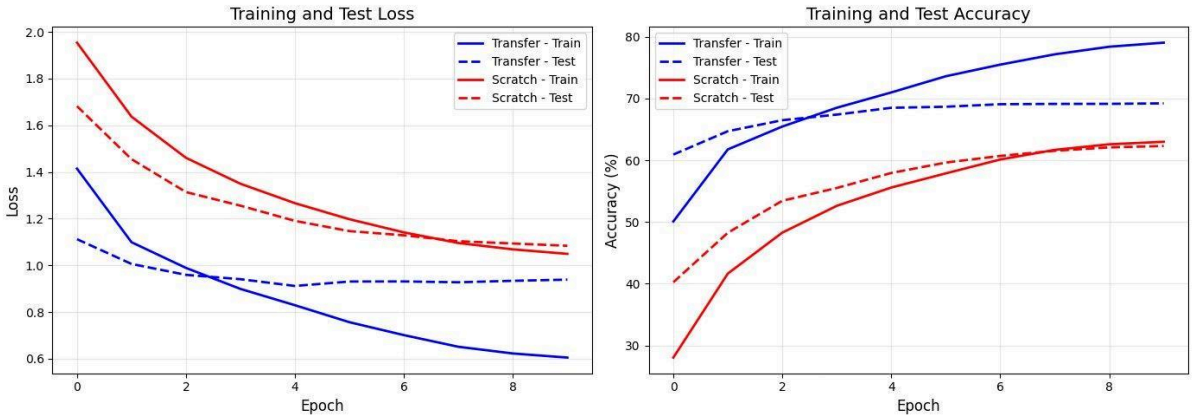

Fig. 2. Loss (left) and accuracy (right) curves over ten epochs. Blue: transfer learning. Red: from scratch. Solid: training set. Dashed: test set. The initial 20.67-point test accuracy gap narrows to 6.90 points by epoch 10.

TABLE III TEST ACCURACY PER EPOCH

Epoch	Transfer Acc.	Scratch Acc.	Gap
1	60.92%	40.25%	+20.67 pp
2	64.70%	48.26%	+16.44 pp
3	66.48%	53.44%	+13.04 pp
4	67.39%	55.51%	+11.88 pp
5	68.49%	57.94%	+10.55 pp
6	68.66%	59.62%	+9.04 pp
7	69.07%	60.70%	+8.37 pp
8	69.12%	61.53%	+7.59 pp
9	69.13%	62.08%	+7.05 pp
10	69.21%	62.31%	+6.90 pp

C. Per-Class Breakdown

Transfer learning beat the from-scratch baseline in all ten classes (Figure 3, Table IV). The gaps ranged from 3.30 points (ship) to 14.80 points (cat). The pattern makes sense: ships, trucks, and automobiles have silhouettes that stay recognisable even after heavy degradation, so the from-scratch model can learn them reasonably well. Cats, birds, and deer look more similar to each other in phosphene form—four limbs, rounded body—and that is where the richer feature set from ImageNet pre-training pays off.

The absolute accuracy numbers also matter. Cats top out at 51.40% even with pre-training; birds at 54.40%. For a real prosthetic system, misclassifying a cat as a dog 50% of the time may be harmless, but misidentifying a car as a person is not. Any deployed system would need either better features for these hard categories or a confidence threshold that withholds a label when the model is uncertain.

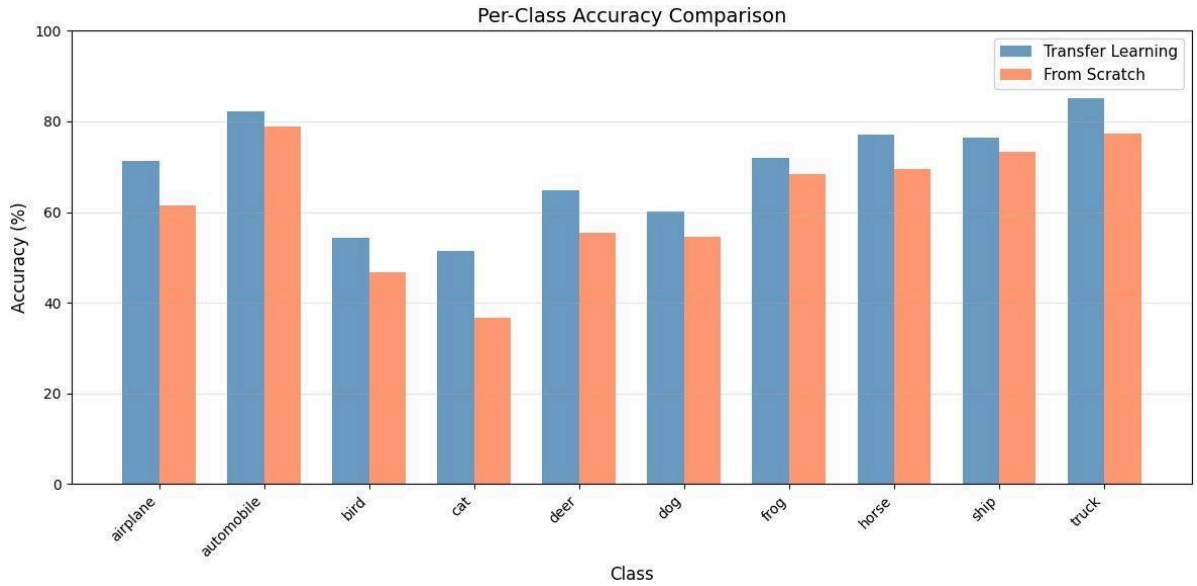

Fig. 3. Per-class test accuracy: transfer learning (blue) vs. from scratch (orange). Transfer learning leads in every class. The largest gaps are in animal categories where phosphene silhouettes are similar across classes.

TABLE IV PER-CLASS TEST ACCURACY

Class	Transfer	Scratch	Improvement
Airplane	71.20%	61.50%	+9.70 pp
Automobile	82.20%	78.80%	+3.40 pp
Bird	54.40%	46.80%	+7.60 pp
Cat	51.40%	36.60%	+14.80 pp
Deer	64.80%	55.40%	+9.40 pp
Dog	60.20%	54.50%	+5.70 pp
Frog	71.90%	68.40%	+3.50 pp
Horse	77.00%	69.60%	+7.40 pp
Ship	76.50%	73.20%	+3.30 pp
Truck	85.20%	77.40%	+7.80 pp
Mean	69.48%	62.16%	+7.32 pp

V. DISCUSSION

A. What the Epoch-1 Gap Tells Us

The 20.67-point gap at epoch 1 is the most informative number in the results. At that point, the pre-trained model has seen the phosphene training data for exactly one pass and has barely had time to adjust its weights. The from-scratch model has had the same one pass but started from nothing. The fact that the pre-trained model is already 20 points ahead says that ImageNet weights, as-is, encode something useful for this task.

The explanation is probably in the early layers. Edge and boundary detectors are among the first features that form during ImageNet training [11], and they are also the features that survive phosphene simulation. Colour is gone. Texture is gone. What remains is roughly where object edges were—which is exactly what those early-layer filters respond to. The from-scratch model can learn the same filters eventually, and by epoch 10 it mostly has, but it needed ten full epochs of phosphene data to get there.

The data-efficiency angle matters for real deployments. You cannot ask prosthesis users to provide labelled training data at scale. The cost and ethics of that kind of data collection are prohibitive. Starting from ImageNet weights cuts the amount of phosphene-specific training required, which makes it a practical as well as a theoretical advantage.

B. What This Means for Prosthesis Design

The direct takeaway is that the image-processing module of a cortical prosthesis should start from pre-trained weights rather than being trained from scratch on phosphene data alone. EfficientNet-B0 is small enough to run on embedded hardware, so the pre-training approach does not conflict with the power or latency requirements of a wearable device.

The per-class results point toward a second design choice. Categories with low accuracy even after pre-training—birds (54%) and cats (51%)—are the ones where phosphene images are most ambiguous. A hierarchical classifier that first separates animals from vehicles before attempting fine-grained recognition would avoid the worst errors. Alternatively, the system could output a confidence score and withhold a label below a threshold, which is probably safer than a low-confidence guess.

C. Limitations

Three limitations are worth stating plainly. First, CIFAR-10 is an easy dataset by current standards: ten classes, centred objects, clean backgrounds. Real prosthetic scenes are cluttered, multi-object, and scale-variable. Second, our phosphene simulator is a rough approximation. A real implant produces phosphenes whose shape, size, and stability depend on per-electrode spread functions, cortical magnification, and individual anatomy—none of which we model. Third, ten epochs is a short run. The from-scratch model was still improving at epoch 10, so the final gap may shrink with longer training. That said, the pre-trained model had already plateaued, so the relative advantage would likely hold.

VI. FUTURE WORK

A few extensions seem worth pursuing:

- **Architecture comparison** — we only tested EfficientNet-B0. ResNet-50 and Vision Transformer variants might respond differently to phosphene degradation, and larger EfficientNet models might close the accuracy gap on the hard classes.
- **Electrode count ablation** — varying the downsampling grid from 16 × 16 to 128 × 128 would show how the transfer advantage changes as electrode count increases, which has direct implications for implant design.
- **More realistic images** — CIFAR-10's centred, single-object images are not representative of real scenes. Testing on COCO or an ImageNet subset would give a more honest picture.
- **Self-supervised pre-training on phosphene data** — rather than starting from natural-image weights, one could pre-train on a large corpus of unlabelled phosphene images. This might close the domain gap further.
- **Validation with actual users** — the simulation needs to be checked against perceptual reports from prosthesis users. If the simulator does not capture what users actually see, the results here may not generalise.

VII. CONCLUSION

We compared transfer learning from ImageNet against random initialisation for object recognition in phosphene-simulated images, using EfficientNet-B0 and CIFAR-10. The pre-trained model finished at 69.21% test accuracy; the from-scratch model finished at 62.31%. The 6.90-point final gap understates the advantage: at epoch 1, before much fine-tuning had happened, the pre-trained model led by over 20 points. That early gap suggests the benefit comes from ImageNet edge detectors that happen to match the kind of structural information that survives phosphene simulation.

For anyone building the processing layer of a cortical visual prosthesis, the implication is straightforward. Start from pre-trained weights. EfficientNet-B0 is small enough to deploy on wearable hardware, the accuracy gain is real, and the data-efficiency advantage matters when labelled prosthetic training data is hard to collect. The simulation code and training scripts from this study are publicly available.

VIII. REFERENCES

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